

MEMORANDUM - Overview of Model Cards for Responsible AI

To: Justin Bullock, Americans For Responsible Innovation

From: Saumya Goyal, Vashishth Doshi, Ziqian Zhong

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Executive Summary

Model cards are standardized documentation attached to AI systems that summarize how a model was built, what it was tested on, and where it may fall short. Originally proposed by [Mitchell et al. 2019] to prevent harms like demographic bias from going undetected until after deployment, they have since been updated for large language models and echoed by parallel transparency efforts across medicine, data science, and linguistics. This memo explains what model cards are, who they serve, and why they represent a necessary foundation for responsible AI deployment.

Need for Model Cards

Machine learning (ML) models are software systems trained on large amounts of data to perform tasks like generating text, making predictions, or flagging content. As these systems have been deployed in consequential domains — hiring, lending, healthcare, criminal justice — a recurring problem emerged: issues like racial and demographic bias were only being discovered after the fact, once real harm had already occurred. The root cause was a lack of any consistent standard for documenting how models were built, what they were tested on, and what they were designed to do.

Model cards were introduced by [Mitchell et al. 2019] as a solution to this problem. A model card is a standardized document that travels with an AI model and summarizes its purpose, performance characteristics, data sources, and known limitations as a kind of nutritional label for AI systems. The goal is to make models legible before deployment, not just after something goes wrong.

Goals for Model Cards

Model cards are designed to serve multiple audiences simultaneously. Developers building on top of existing models can make better implementation decisions when they understand what a model was trained on and where it underperforms. Organizations and policymakers evaluating whether to adopt an AI system can assess its suitability for their specific context. And individuals subject to AI-driven decisions — an automated hiring screen, a loan approval algorithm — can better understand what shaped the outcome. Core elements of a standard model card typically include intended use cases, performance metrics across different demographic groups, and details about training data provenance.

Model Cards in Literature

Related approaches have emerged in parallel across other fields. TRIPOD [Collins et al. 2015] establishes reporting standards for clinical prediction models in medicine. Dataset Nutrition Labels [Holland 2018] apply a similar transparency logic to the datasets that underlie AI

systems. Data Statements for NLP[Bender et al. 2018] require structured documentation of the linguistic data used to train natural language processing models. These parallel efforts reflect a broader consensus across technical communities that transparency documentation is a precondition for responsible deployment.

As large language models (LLMs) have become the dominant form of AI system in public use, the original model card framework has been updated to address their specific characteristics. [Mamirov et al. 2025] propose an LLM-era extension that adds dataset size, environmental impact, and explicit safety evaluations to the model card template, reflecting the scale, energy intensity, and higher-stakes deployment contexts that define frontier AI systems today.

Model cards are not a complete solution to the challenges of AI accountability, but they are a necessary foundation for it. Without consistent, structured documentation of how AI systems are built and where they fall short, neither regulators, nor deployers, nor the public have the information needed to make informed decisions about their use.